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June 8, 2026

Editorial Office
Physical Review A
American Physical Society

Dear Editors,

We submit the accompanying manuscript, “Which quantum algorithms suit photonic continuous-variable hardware? A comparative resource and noise-robustness study,” for consideration as a Regular Article in *Physical Review A*. The work falls squarely within the journal’s scope of quantum information, quantum computation, and the physics of quantum systems.

Motivation. Photonic continuous-variable (CV) architectures support deterministic, room-temperature generation of large cluster states and are a leading route to fault-tolerant quantum computing, where overhead is dominated by squeezing and by the consumption of non-Gaussian resource states. Different algorithms place very different demands on these resources, yet no unified comparison has quantified which algorithm classes are intrinsically robust on CV hardware and how much Gottesman–Kitaev–Preskill (GKP) error correction actually helps.

Main results. We port four representative algorithms—control-free phase estimation, randomized product formulas, quantum Kolmogorov–Arnold networks, and Regev factoring—to a common CV model and compare them under photon loss (exact Fock-space Kraus channel) and Gaussian displacement with GKP correction. We establish: (i) control-free phase estimation is the most noise-robust because it needs no controlled unitaries and, for quadratic Hamiltonians, no non-Gaussian gates, a property that survives for non-Gaussian Hamiltonians; (ii) GKP correction reduces residual error only above the finite-squeezing floor (proved); whether an algorithm benefits is set by a noise-versus systematic-limited dichotomy whose empirical envelope constants we report with $R^2 \geq 0.89$; (iii) the comb readout of Regev requires squeezing scaling as $\sqrt{n}$ (proved), demonstrating that CV-native cheapness of state preparation and the Fourier transform does not rescue a depth- and readout-heavy algorithm; (iv) against a qubit surface-code baseline, the CV advantage is structural—Gaussian operations are free—rather than threshold-based.

Methodological rigor. The core GKP-correction surrogate is validated against an explicit Glancy–Knill error-correction Monte Carlo to within 0.1%, lifting a common concern in CV-resource analyses. A two-dimensional noise phase diagram and Hamiltonian ensembles confirm the predictions are not artefacts of single operating points or single instances. We are explicit about which results are exact, which are validated model surrogates, and which are extrapolations.

New worked physics examples. Responding to earlier feedback that the work lacked concrete physics results, the manuscript now includes quantitative demonstrations on *three diatomic molecules*— H_2 , HF , and I_2 , spanning a decade in vibrational quantum ($\omega_e = 214\text{--}4401\text{ cm}^{-1}$) and an order of magnitude in anharmonicity ($x_e = 0.003\text{--}0.028$)—using experimentally fixed Huber–Herzberg Morse parameters (Sec. VIII). CV control-free QPE recovers the populated vibrational

eigenenergies of all three molecules to within 0.55 cm^{-1} *below the standard 1 cm^{-1} spectroscopic threshold*; weakly anharmonic I_2 reaches $\sim 10^{-2} \text{ cm}^{-1}$. We further show that matching this accuracy with a discrete-variable QPE requires $\sim 1.5 \times 10^{10}$ T gates and $\sim 16,000$ controlled time evolutions, whereas the CV control-free protocol needs zero controlled unitaries and zero non-Gaussian gates for the harmonic part. This is, to our knowledge, the first end-to-end CV-resource-quantified calculation of *multiple* real molecular spectra at spectroscopic accuracy, and we believe it constitutes a meaningful physics contribution beyond methodology. A Chebyshev-KAN classifier trained on the recovered eigenstates assigns the vibrational quantum number with 100% accuracy under realistic measurement noise.

Submission history. An earlier short-form version of this work was returned by *Applied Physics Express* on scope grounds (the referee judged the topic to lie in fundamental quantum-information physics rather than applied photonics) and a preliminary version of the present manuscript was returned by PRA on the grounds that the original content did not contain significant new physics results. The H_2 vibrational-spectrum calculation reported in Sec. VIII directly addresses that concern, providing a concrete, quantitative physics deliverable while retaining the original cross-algorithm methodology. The manuscript has not been published elsewhere and is not under consideration at any other journal; the authors declare no competing interests. All scripts and numerical results that reproduce every figure and table are available with the submission.

We thank you for your consideration.

Sincerely,

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